

Notes and Ideas

(parameter-wise functional sensitivity paper)

1 Split the central object: g vs. δ

Situation. §2 uses $g = Xf_\theta$, to answer two different questions and §4 draws a conclusion (“perturbing these parameters contributes more strongly to symmetry breaking”) which **I am not sure g supports.** Define the group average and its complement,

$$\Pi f = \frac{1}{2\pi} \int_0^{2\pi} \rho(h_\varphi)^{-1} f(h_\varphi \cdot x) d\varphi, \quad Q = I - \Pi, \quad \delta = Qf_\theta,$$

so δ is the non-invariant part of the learned function. Working in $\mathcal{H} = L^2(\mu; \mathbb{R}^m)$ with a **G -invariant probe** measure μ , the **action is unitary and X is skew-adjoint**, hence:

Proposition 1. **$\langle \delta, g \rangle = \langle \delta, X\delta \rangle = 0$.**

So g is *tangent to the group orbit of f_θ* : moving along it rotates the learned function’s “phase” at *constant* distance from the equivariant subspace $\mathcal{E} = \ker X$. **It does not break or restore symmetry at first order.**

Idea. Keep both objects, each with its own attribution — same J_θ^+ , different right-hand side, i.e. one extra line of code:

object	meaning	attribution answers
$g = Xf_\theta$	orbit velocity	which parameters <i>represent/carry</i> the symmetry (current Fig. 1a — keep, relabel)
$\delta = Qf_\theta$	departure from $\mathcal{E}$	which parameters <i>control the defect</i> — licenses the “breaking” language

Two sets of c_i coeffs then?

State Proposition 1 in §2 with its three-line proof. Prediction to check empirically: the **two attributions share support and agree at module level, but differ on within-module rankings.**

This also gives §5’s **“equivariance residual”** a definition: report $\|P_{T^\perp} g\| / \|g\|$ and $\|P_{T^\perp} \delta\| / \|\delta\|$ per model.

2 Scale of the c_i (Fig. 1a) and the L_1 confound

Situation A — global scale. $c = J_\theta^+ g$ is linear in g . As the ASRNN becomes near-invariant, $\|g\|$ also decreases, so *every* $|c_i|$ drops. This **could confound with the behaviour in §4.**

Only confounds if scaling of lc_il occurs non-uniformly

How does below solve this?

Idea A. Solve against $\hat{g} = g / \|g\|$; plot shares $|c_i| / \sum_j |c_j|$ (init vs. trained); report per checkpoint:

$$\underbrace{\|g\|}_{\text{order parameter}}, \quad \underbrace{\|P_T g\| / \|g\|}_{\text{representation quality}}, \quad \underbrace{\text{PR}(c) = \left(\sum_i |c_i| \right)^2 / \sum_i c_i^2}_{\text{concentration}}.$$

Situation B — per-parameter scale. §6.1 adds L_1 regularization “to bias training towards sparser parameter representations” — and §4’s headline finding is a sparse set of dominant c_i . Each c_i has units $1/\theta_i$, so L_1 -shrunk parameters are systematically distorted; the two-cluster structure may just be the L_1 support.

Motivating sensitivity normalisation!!!!? What is the effect

Idea B (diagnostics first, no retraining).

1. Spearman $\rho(|c_i|, \|S_i\|)$ and $\rho(|c_i|, |\theta_i|)$ — high values $\Rightarrow c$ is reporting the L_1 support.
2. Check whether the top- $|c_i|$ cluster coincides with the L_1 non-zeros.
3. Use the scale-invariant score

$$\tilde{a}_i = \frac{c_i \langle S_i, g \rangle}{\|P_T g\|^2}, \quad \sum_i \tilde{a}_i = 1,$$

invariant under per-parameter rescaling $\theta_i \mapsto \lambda_i \theta_i$, signed, and Lorenz-curve-able (“top 10% carry $X\%$ ”).

How is this different to sensitivity normalisation?

Function-space geometry: $c^{\text{rep}} = J^+ g$ moves f_θ along the orbit;
 $c^{\text{def}} = J^+ \delta$ moves it toward/away from $\mathcal{E}$

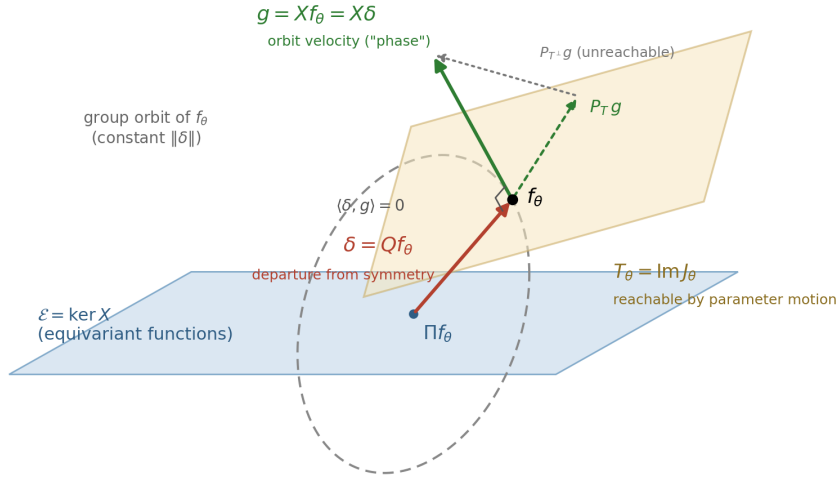


Figure 1: f_θ sits on its group orbit (dashed) at distance $\|\delta\|$ from $\mathcal{E}$ (blue). g is tangent to the orbit, δ is radial, and they are orthogonal. T_θ (tan) is what parameter motion can reach; $\|P_{T^\perp} g\| / \|g\|$ is the (currently unreported) representation residual.

Null control (one extra solve; put it in the paper). $J_\theta^+ g_{\text{rand}}$ for a random matched-norm target. If it yields the same concentration/Lorenz curve, the structure is the conditioning of J (Σ_r^{-1} amplifying small singular directions), not symmetry. Plot as a dashed null curve.

We have plots of the relative projection error